Biplanes: Game Design Document

Juan Galeazzi: Final Draft 10/25/2026

1. **Executive Summary**

Biplanes is a side scrolling WWI airplane shooter where the player tries to survive waves of enemy airplanes and ground fire, while navigating through a sky full of obstacles like barrage balloons, mountainous terrains and rural buildings before finally encountering the mission objective.

- Win Conditions: player score points by destroying enemies and completing it stated missions.
- Loss State: Player biplane gets damaged every time it gets hit by enemy fire until it finally crashes. Player biplanes also crash when colliding into terrain or buildings.

Visual Style: Blueprints, the game will use a duotone style with wireframes to display the main elements (hero plane, enemies, targets) while the background elements will be more loosely drawn like with pencil illustration style).

2. **Game Mechanics**

Core Loop: Player controls the airplane up and down and left to right to shoot enemies and avoid enemy fire, terrain and buildings.

Scoring: Enemy kills add points to the score, the harder the enemy the more points awarded.

Milestones/Save Progress: The game will be divided in 5 missions each one will be of increasing difficulty.

Player Airplane: A WWI era biplane with a forward-facing machine gun, and seven bombs capacity for ground attacks. Player airplane will be identifiable by its bullseye marking on the center of its body

Enemies:

1. Airplanes: Enemy airplanes will be identifiable by the Iron Cross on the center of its body and tail. They will shoot at player airplane and only can be destroyed with direct hits.

2. Anti-Aircraft Artillery: They fire from the ground up to try and hit the player airplane, they must be destroyed by the player by dropping bombs.

3. Bombers: Bombers are bigger airplanes they usually have a fighter escort, and they are trying to drop bombs into friendly targets. They must take multiple hits to be destroyed.

4. Zeppelins: Big Aircrafts with multiple flying positions player must evade its fire while disabling the Zeppelins turrets by hitting them. Player has to hit the Zeppelin several times to destroy it.

Obstacles

1. Barrage Balloons: Low flying balloons that must be destroyed by player if the mission requires it, usually protected by AAA
2. Buildings and Trees, if flying too low player could crash into them.

Airplane damage: Airplane damage increases with each hit when damage reaches 100% the airplane is destroyed. When Damage is about 70% trail smoke will be shown behind the plane. Health Ballons will show up on some points in the map, navigating through them will replenish health.

Player Airplane Controls: keyboard using WASD keys for direction, space bar for fire and Enter key for dropping bombs.

Missions:

The game will consist of five different missions with incremental difficulty:

Mission one: Destroy incoming airplanes- Player starts at airfield where a wave of enemy airplanes is inbound. Player must takeoff and destroy all incoming enemy airplanes.

Mission two: Destroy enemy bomber - An enemy bomber is approaching the airfield escorted by fighter planes. Destroy the bomber before it reaches its target and drops its bomb.

Mission three: Destroy enemy defenses - Go into enemy territory and destroy AAA using bombs, while also evading enemy fire from ground and from enemy airplanes.

Mission 4: Destroy enemy base - Go deeper into enemy territory destroying barrage balloons, AAA's and enemy planes to finally deliver a bomb right into the enemy's base.

Mission 5: Enemy Zeppelin – The enemy is deploying its newest more power weapon, a massive Zeppelin equipped with anti-aircraft turrets and escorted by enemy fighters. Player must destroy it before it reaches its destination (the player's HQ)

3. **AI Architecture**

Art Agent: Creates all the games visual elements. Establish game style and aesthetics.

Game Mechanics Agent: Construct core framework logic, physic and collisions handlers.

Orchestration Agent: break down high level game concept into specialized roles,

manage GDD, architecture UI and QA strategy.

Mission Design Agent: responsible for crafting each of the five missions.

QA Agent: make sure that game is playable on intended platform / controls are accurate and game is free of bugs.

4. Technical strategy

Deployment:
Unity Engine: initial release playable on PC/Mac and Web.

Key Constraints:

Use of unknow engine (Unity) may require a steeper learning curve.
Six weeks limited time and solo development.

Token Budget: AI Agents will run on a local Nvidia Spark DGX machine running opencode with Qwen3.6: 35b , therefore no budget is necessary, however the downsize is that it won't be using the latest frontier models and the processing time will be longer.